

Unit I

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Computer systems

A Computer is an electronic device which performs operations such as accepts data as an input, store the data, manipulate or process the data and produce the result as an output.

Main task performed by a computer

- Accept the data
- Process or manipulate the data
- Display or store the result in the form of human understanding
- Store the data, instructions and results.

Characteristics of Computers

Speed

You will be surprised to know that computer can perform millions (1,000,000) of instructions and even more per second. Therefore, we determine the speed of computer in terms of microsecond (10^{-6} part of a second) or nano-second (10^{-9} part of a second). From this you can imagine how fast your computer performs work.

Accuracy

Suppose someone calculates faster but commits a lot of errors in computing. Such result is useless. There is another aspect. Suppose you want to divide 15 by 7. You may work out up to 2 decimal places and say the dividend is 2.14. I may calculate up to 4 decimal places and say that the result is 2.1428. Someone else may go up to 9 decimal places and say the result is 2.142857143. Hence, in addition to speed, the computer should have accuracy or correctness in computing. The degree of accuracy of computer is very high and every calculation is performed with the same accuracy. The accuracy level is determined on the basis of design of computer. The errors in computer are due to human and inaccurate data.

Diligence

A computer is free from tiredness, lack of concentration, fatigue, etc. It can work for hours without creating any error. If millions of calculations are to be performed, a computer will perform every calculation with the same accuracy. Due to this capability it overpowers human being in routine type of work.

Versatility

It means the capacity to perform completely different type of work. You may use your computer to prepare payroll slips. Next moment you may use it for inventory management or to prepare electric bills.

Power of Remembering

Computer has the power of storing any amount of information or data. Any information can be stored and recalled as long as you require it, for any numbers of years. It depends entirely upon you how much data you want to store in a computer and when to lose or retrieve these data.

No IQ

Computer is a dumb machine and it cannot do any work without instruction from the user. It performs the instructions at tremendous speed and with accuracy. It is you to decide what you want to do and in what sequence. So a computer cannot take its own decision as you can.

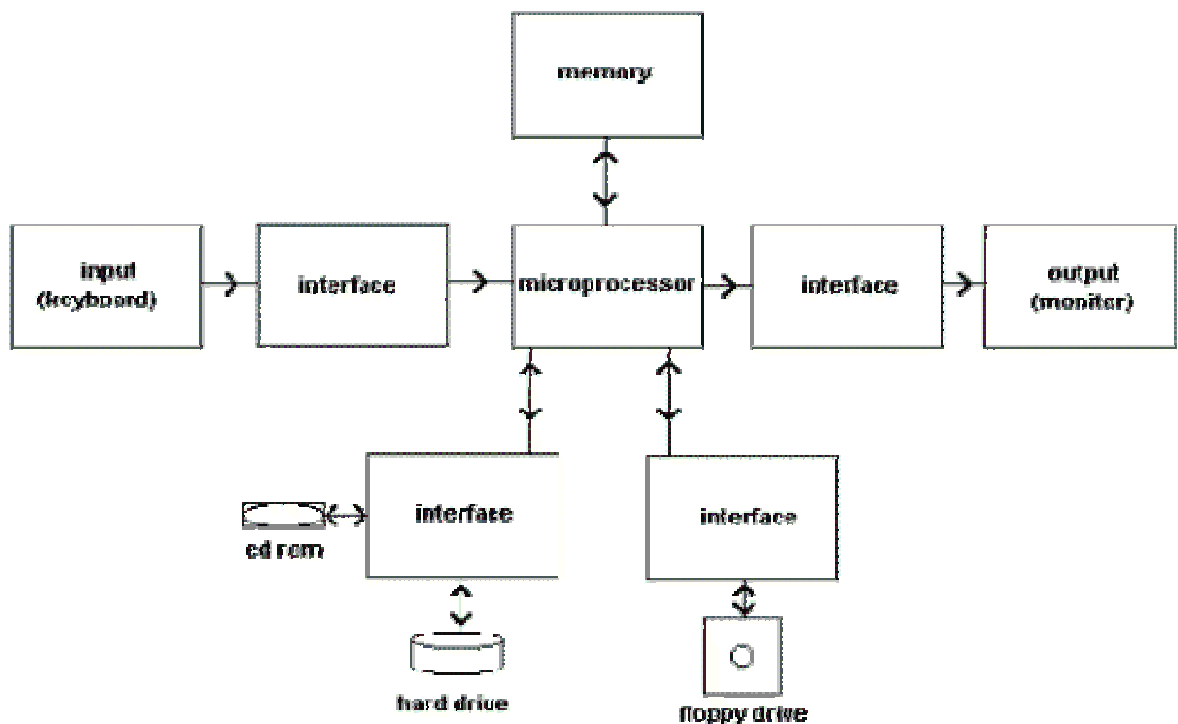
No Feeling

It does not have feelings or emotion, taste, knowledge and experience. Thus it does not get tired even after long hours of work. It does not distinguish between users.

Storage

The Computer has an in-built memory where it can store a large amount of data. You can also store data in secondary storage devices such as floppies, which can be kept outside your computer and can be carried to other computers.

Block Diagram of Computer



Hardware: The physical components which are used to assemble computer system are known as hardware. Any computer device which we can touch including cables and wires is hardware.

Input Unit:

Computers need to receive data and instruction in order to solve any problem. Therefore we need to input the data and instructions into the computers. The input unit consists of one or more input devices. Keyboard is the one of the most commonly used input device. Other commonly used input devices are the mouse, floppy disk drive, magnetic tape, etc. All the input devices perform the following functions.

- Accept the data and instructions from the outside world.
- Convert it to a form that the computer can understand.
- Supply the converted data to the computer system for further processing.

Storage Unit:

The storage unit of the computer holds data and instructions that are entered through the input unit, before they are processed. It preserves the intermediate and final results before these are sent to the output devices. It also saves the data for the later use. The various storage devices of a computer system are divided into two categories.

1. **Primary Storage:** Stores and provides very fast. This memory is generally used to hold the program being currently executed in the computer, the data being received from the input unit, the intermediate and final results of the program. The primary memory is temporary in nature. The data is lost, when the computer is switched off. In order to store the data permanently, the data has to be transferred to the secondary memory.
2. **The cost of the primary storage is more compared to the secondary storage.** Therefore most computers have limited primary storage capacity. **Secondary Storage:** Secondary storage is used like an archive. It stores several programs, documents, data bases etc. The programs that you run on the computer are first transferred to the primary memory before it is actually run. Whenever the results are saved, again they get stored in the secondary memory. The secondary memory is slower and cheaper than the primary memory. Some of the commonly used secondary memory devices are Hard disk, CD, etc.,

Memory Size:

All digital computers use the binary system, i.e. 0's and 1's. Each character or a number is represented by an 8 bit code. The set of 8 bits is called a byte. A character occupies 1 byte space. A numeric occupies 2 byte space. Byte is the space occupied in the memory.

The size of the primary storage is specified in KB (Kilobytes) or MB (Megabyte). One KB is equal to 1024 bytes and one MB is equal to 1000KB. The size of the primary storage in a typical PC usually starts at 16MB. PCs having 32 MB, 48MB, 128 MB, 256MB memory are quite common.

Output Unit:

The output unit of a computer provides the information and results of a computation to outside world. Printers, Visual Display Unit (VDU) are the commonly used output devices. Other commonly used output devices are floppy disk drive, hard disk drive, and magnetic tape drive.

Arithmetic Logical Unit:

All calculations are performed in the Arithmetic Logic Unit (ALU) of the computer. It also does comparison and takes decision. The ALU can perform basic operations such as addition, subtraction, multiplication, division, etc and does logic operations viz, >, <, =, 'etc. Whenever calculations are required, the control unit transfers the data from storage unit to ALU once the computations are done, the results are transferred to the storage unit by the control unit and then it is send to the output unit for displaying results.

Control Unit:

It controls all other units in the computer. The control unit instructs the input unit, where to store the data after receiving it from the user. It controls the flow of data and instructions from the storage unit to ALU. It also controls the flow of results from the ALU to the storage unit. The control unit is generally referred as the central nervous system of the computer that control and synchronizes its working.

Central Processing Unit:

The control unit and ALU of the computer are together known as the Central Processing Unit (CPU). The CPU is like brain performs the following functions:

- It performs all calculations.
- It takes all decisions.
- It controls all units of the computer.

A PC may have CPU-IC such as Intel 8088, 80286, 80386, 80486, Celeron, Pentium, Pentium Pro, Pentium II, Pentium III, Pentium IV, Dual Core, and AMD etc.

Software

Software is a set of program or instructions given to the computer to carry out certain jobs or tasks. It is set of programs which actually makes the computer useful. The software can be classified into three categories

System Software: System software is responsible for managing a variety of independent hardware components, so that they can work together. Its purpose is to unburden the application software programmer from the often complex details of the particular computer being used, including such accessories as communications devices, printers, device readers, displays and keyboards, and also to partition the computer's resources such as memory and processor time in a safe and stable manner.

- Device drivers
- Operating systems
- Servers
- Utilities
- Window systems

Programming Software: Programming Software usually provides tools to assist a programmer in writing computer programs, and software using different programming languages in a more convenient way. The tools include:

- Compilers
- Debuggers
- Interpreters
- Linkers
- Text editors

Application Software: Application software is developed to aid in any task that benefits from computation. It is a broad category, and encompasses Software of many kinds, including the internet browser being used to display this page. This category includes:

- Business software
- Computer aided design
- Databases
- Decision making software
- Educational software
- Image editing
- Industrial automation
- Mathematical software
- Medical software
- Simulation
- Spreadsheets
- Telecommunications
- Video games
- Word processing

Applications of Computers

Computers applications are used worldwide in almost all fields.

1. Science: scientists use computers to analyze large quantity of data to analyze the effect of earthquake on buildings dams, for satellite launching and remote controlling etc.
2. Education: Computers are used in all educational institutions ranging from ordinary schools to universities for variety of purpose like training, e-learning, exams and results.
3. Medicine and Health: The doctors and technicians use computer to diagnose the human problems and monitor them to provide treatment.
4. Engineering/ Architecture / Manufacturing: They are used for planning, designing, and drawing the layouts.
5. Entertainment: Computers used to listen the music, play the video games, graphics in movies.
6. Communication: to send and receive mails, e-commerce, e-shopping, e-banking and e-learning.
7. Business and Banking Applications: Computers are used in all commercial and business organizations.
8. Government: Used for weather forecasting, in military, satellite launching and controlling, communication and e-governance.

Computing Environments

Personal Computing Environment

All of the computer hardware components are tied together in our personal computer. A **personal computer (PC)** is a computer whose original sales price, size, and capabilities make it useful for individuals, and intended to be operated directly by an end user, with no intervening computer operator. People generally relate this term with Microsoft's Windows Operating system. Personal computers generally run on Windows, Mac or some version of Linux operating system.

Desktop: Desktop computer is just another version of Personal Computer intended for regular use from a single user. A computer that can be fit on a desk can also be called as desktop.

Time-Sharing Environment

In the time-sharing environment, all computing must be done by the central computer. The central computer the shared resources, it manage the shared data and printing. Employees in large companies often work in what is known as time-

sharing environment. In the time sharing environment, many users are connected to one or more computers. These computers may be mini computers and central mainframes. In this environment the output devices, auxiliary storage devices are shared by all the users.

Client/Server Environment

Client/Server computing environment splits the computing function between a central computer and user's computers. The users are given personal computers or work stations so that some of the computation responsibility can be moved from the central computer and assigned to the workstations. In the client/server environment the users micro computers or work stations are called the client. The central computer which may be a powerful micro computer, minicomputer or central mainframe system is known as server.

Distributed Computing Environment

A distributed computing environment provides a seamless integration of computing functions between different servers and clients. The internet provides connectivity to different servers throughout the world. This environment provides reliable, scalable and highly available network.

Computer Languages

Programming language is used to communicate user with the computer give ideas, logics, thoughts etc. These are used to develop programs. These are classified as:

1. Low Level Languages
2. High Level Languages

Low Level Languages:

1. Machine / Binary Language: It is a first generation programming language. It is the language based on binary digits (0's and 1's).
 - a. Advantages:
 - i. Computer can understand directly.
 - b. Disadvantages:
 - i. It is very difficult to remember the codes and address of memory locations.
 - ii. User can't modify the program.
 - iii. User can't debug the program.
 - iv. It is machine dependent.
 - v. It is suitable for simple applications.
2. Assembly / Symbolic Language: It is a second generation programming language. This language based on symbolic codes (mnemonic) to represent any instruction.
 - a. Advantages:

- i. User can remember the mnemonics.
 - ii. It is easy to understand and develop the programs.
 - iii. User can modify the program and debug.
 - iv. It is suitable for simple applications.
 - b. Disadvantages:
 - i. It is machine dependent.
 - ii. It requires the translator program called Assembler.
3. High Level Language: It is a third generation programming language. It is based on English grammatical notations and mathematical formulae's. these are classified as general and specific purpose language.
- a. Advantages:
 - i. Easy to follow.
 - ii. Easy to understand
 - iii. Easy to modify and debug.
 - iv. Suitable for complex applications.
 - b. Disadvantages:
 - i. It requires the translator program called Compiler or Interpreter.
 - ii. It runs programs slower with compare to low level languages

Creating and Running Programs

There are four steps in this process.

1. Writing and editing the program using Text editor (source code).
2. Compile the program using any C compiler.(.bak file)
3. Linking the program with the required library modules(object file)
4. Executing the program. (.Exe file)

Creating and Editing a C Program in C Programming Language compiler:

Writing or creating and editing source program is a first step in c language. *Source code* is written in *c programming language* according to the type of problem or requirement, in any text editor.

Saving C Program in C Programming Language: *Source code* is saved on the secondary storage. Source code is saved as text file. The extension of file must be ".c". Example the file name is "*learn c programming language.c*"

Compiling C program in C Programming Language: Computer does not understand *c programming language*. It understands only 0 and 1 means *machine language*. So *c programming language code* is converted into *machine language*. The process of converting *source code* in to *machine code* is called compiling. Compiler is a program that *compiles source code*. Compiler also *detects errors* in

source program. If compiling is successful *source program* is converted into *object program*. Object program is saved on disk. The extension of file is ".obj"

Linking in C programming Language: There are many built in functions available in *c programming language*. These functions are also called *library functions*. These *functions* are stored in different *header files*.

Loading program: The process of transferring a program from secondary storage to main memory for execution is called *loading a program*. A *program* called loader does loading.

Executing program: Execution is the last step. In this *step program* starts execution. Its instructions start working and output of the program display on the screen.

Program Development Method

1. Specifying and analyzing the problem statement.
2. Designing an Algorithm
3. Coding and Implementation
4. Debugging
5. Testing and validating
6. Documentation and maintenance

Specifying and analyzing the problem statement: the problem which has to be implemented into a program must be thoroughly understood before the program is written. Problem must be analyzed to determine the input and output requirements of the program. A problem statement is created with these specifications.

Designing an Algorithm: with the problem statement obtained in the previous step, various methods available for obtaining the required solution are analyzed and the best method is designed into algorithm.

Coding and implementation: the actual problem is written in the required programming language with the help of information depicted in flow charts and algorithms.

Debugging: there is a possibility of occurrence of errors in programs. These errors must be removed to ensure proper working of programs. Hence solving the program without errors is known as debugging.

Types of errors that may occur in the program are:

- a) Syntactic Errors(Compilation Errors): These errors occur due to the usage of wrong syntax for the statements.
- b) Runtime Errors: These errors are determined at the execution time of the program.

- c) Logical Errors: These errors occur due to incorrect usage of the instructions in the program.

Testing and Validating: Testing and Validation is performed to check whether the program is producing correct results or not for different values according to user requirement.

Documentation and Maintenance: Documentation is the process of collecting, organizing and maintaining, in written the complete information of the program for future references. Maintenance is the process of upgrading the program according to the changing requirements.

Algorithm / Pseudo Code

Algorithm is a method of representing the step by step logical procedure for solving a problem. It is a tool for finding the logic of a problem.

Algorithm Properties:

1. Finiteness: an algorithm must terminate in a finite number of steps.
2. Definiteness: Each step of an algorithm must be clear and easy to understand (unambiguous).
3. Effectiveness: Each step must be effective, in the sense that should be primitive (easily convertible to program).
4. Generality: The algorithm must be complete in itself so that it can be used to solve all problems of a specific type for any input data.
5. Input / Output: Each algorithm must take zero, one or more quantities as input data and produce one or more output values.

Writing an algorithm

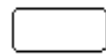
An algorithm can be written in English, like sentences and using mathematical formulas. Sometimes algorithm written in English like language is Pseudo code.

Examples

- 1) Finding the average of three numbers
 1. Let a,b,c are three integers
 2. Let d is float
 3. Display the message "Enter any three integers:"
 4. Read three integers and stores in a,b,c
 5. Compute the $d = (a+b+c)/3.0$
 6. Display "The avg is:" , d
 7. End.

Flow Chart

The pictorial representation of an algorithm using some shapes and symbols is known as flow chart.



Start or end of the program



Computational steps or processing function of a program



Input or output operation



Decision making and branching



Connector or joining of two parts of program



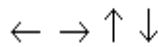
Magnetic Tape



Magnetic Disk



Off-page connector



Flow line

Annotation



Display



The Flowchart Symbols and Their Usage

Terminal points indicate the starting [sometimes known as a 'trigger'] and ending points of a process.

Step represents a single step within a process, and usually contains the name of a specific action.

Page symbols refer to individual web pages, which may or may not contain multiple elements.

File symbols represent those data elements that exist independently of navigational properties outside of that page, e.g., audio sounds, movie clips, or a portable document file (PDF).

A decision point indicates a sequence in the process at which the end user chooses an option, i.e., a "yes-no", or "true-false" response, and then branches to different parts of the flowchart.

Arrows and connecting lines diagram the logical progression through the course, subject to the choices made at decision or action points within the process.

The input/action symbol represents a user response that directs the course flow from that point onwards, i.e., an online test, or questionnaire form.

Represents the choice made by the user from mutually exclusive options, e.g., a student choosing among different lesson plans.

Conditional selector is similar to the conditional branch except that the user has the option to choose from a number of paths that will fulfill the requested conditions, e.g., the results of a search engine request.

Pages that share one or more common aspects, and are functionally identical may be simplified as a rounded corner rectangle, such as an on-line test or feedback form.

Reference is used as a connecting point when the flowchart necessitates using more than one page, or refers to a complicated subroutine that would be impossible to contain on the main flowchart page.

Annotations provide helpful comments or explanations, e.g. denoting the location where an undeveloped new page/process will fit into the navigational flow structure, or notes for specific team members for further development.

Flow references and flow areas are symbols for reusable sequences, such as logging in with a specific user id and password to enter the course or to initiate an on-line

quiz. The flow reference symbol acts as a placeholder for the flow area sequence in the chart in every situation in which it is repeated.

Flow area is used as a flow area, it documents sections that share similar components/repeated steps within that flow, and requires the use of the following two symbols: entry and exit points.

Exit point concludes the subroutine, such as when the proper user id and password are verified, and documents where the user re-enters the master flowchart.

Entry point documents the place within the master flowchart where the process deviates into a subroutine.

Process Flowchart Drawing Guide-lines

There is no one right way to develop a flowchart, but the following guide-lines provide a general structure to follow, whether it's of the overall course navigational process, or at the hand-off phase to the various team members to develop more detailed treatment.

1. Start with a simple one-line description or title of the process being flowcharted , e.g., "How to..."
2. Using a top-down hierarchy, start with a terminal symbol, naming this trigger event, e.g., "User accesses course database..."
3. Connect each successive action step in the logical sequence of events.
4. Reference detailed information through annotations or connectors.
5. Follow the process through to completion, denoted by a labeled end terminal symbol, e.g., "exit course."

A well-developed functional flowchart created in the design phase can save hours of wasted manpower time by ensuring the structure, sequencing and branching decision points in a computer based instructional program, support the course goals and objectives before development.

Whether you are the sole creator wearing many hats, or one of many on the development team, sharing a common visual language will guide the project through its many iterations and development phases throughout the instructional design process.

